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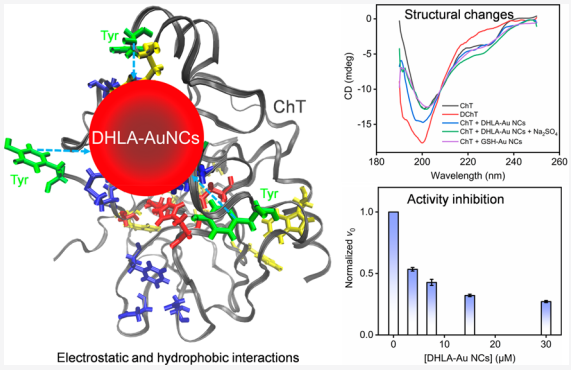
Thermodynamics, Kinetics and Mechanisms of Noncompetitive Allosteric Inhibition of Chymotrypsin by Dihydrolipoic Acid-Coated Gold Nanoclusters

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ABSTRACT: Enzymes are an important class of biomacromolecules which catalyze many metabolic processes in living systems. Nanomaterials can be synthesized with tailored sizes as well as desired surface modifications, thus acting as promising enzyme regulators. Fluorescent gold nanoclusters (AuNCs) are a representative class of ultrasmall nanoparticles (USNPs) with sizes of ∼2 nm, smaller than most of proteins including enzymes. In this work, we chose α-chymotrypsin (ChT) and AuNCs as the model system. Activity assays and inhibition kinetics studies showed that dihydrolipoic acid (DHLA)-coated AuNCs (DHLA-AuNCs) had a high inhibitory potency (IC_{50} = 3.4 μM) and high inhibitory efficacy (>80%) on ChT activity through noncompetitive inhibition mechanism. In distinct contrast, glutathione (GSH)-coated AuNCs (GSH-AuNCs) had no significant inhibition effects. Fluorescence spectroscopy, agarose gel electrophoresis and circular dichroism (CD) spectroscopy were conducted to explore the underlying mechanisms. A two-step interaction model was proposed. First, both DHLA-AuNCs and GSH-AuNCs might be bound to the positively charged sites of ChT through electrostatic forces. Second, further hydrophobic interactions occurred between three tyrosine residues of ChT and the hydrophobic carbon chain of DHLA, leading to a significant structural change thus to deactivate ChT on the allosteric site. On the contrary, no such interactions occurred with GSH of zwitterionic characteristic, which explained no inhibitory effect of GSH-AuNCs on ChT. To the best of our knowledge, this is the first example of the allosteric inhibition of ChT by nano regulators. These findings provide a fundamental basis for the design and development of nano regulators.

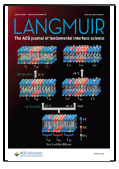


INTRODUCTION

Enzymes are an important class of biomacromolecules which participate in various biochemical reactions in living systems. Meanwhile, enzyme activity should be regulated to an appropriate level, otherwise, it may lead to a series of disorders or diseases.[1−5] Enzyme regulation includes covalent modification and regulator binding.[6] For the latter, regulators can be classified as reversible and irreversible according to the strength of binding. Irreversible regulators permanently bind to the enzymes through covalent bonds, whereas reversible regulators bind enzymes through electrostatic forces, hydrophobic interactions, hydrogen bonds, or van der Waals forces. The most common reversible regulation type is competitive inhibition, where inhibitors bind to the active sites of enzymes, blocking the access of substrates to enzymes. In addition, regulators can also bind to the allosteric sites, leading to the structural changes of enzymes.[7−9] There are many factors affecting enzyme regulation, such as sizes, shapes, and surface charges of both enzymes and regulators, as well as pH, temperature, and ionic strength of the environment.[10,11] Therefore, nanomaterials with tailored sizes and surface modifications are highly attractive as exogenous regulators. These characters allow us to design and synthesize different kinds of nanomaterials according to specific demands.

In the past two decades, nano regulators have been intensively investigated.[12−29] Among them, metal nanoparticles (NPs) and carbon nanomaterials have been widely studied due to their fascinating properties. Rotello et al. have designed and synthesized a series of nano inhibitors based on Au NPs and quantum dots (QDs).[13−18] Dravid et al. demonstrated that graphene oxide competitively inhibited ChT activity with only a slight change in the structure of ChT.[19] Zhou et al. observed that surface functional group density of carbon nanotubes (CNTs) played an important role in enzyme inhibition.[20] Noncompetitive inhibition was induced by CNTs with low

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functionalization density, whereas competitive inhibition was caused by CNTs with high functionalization density. Nicholas et al. discovered that the inhibition of β-galactosidase (GAL) was strongly dependent on the shapes of ZnO NPs. ZnO nanopyramids showed greater inhibition with noncompetitive mechanism; ZnO nanoplates exhibited competitive mechanism; ZnO nanospheres had little effect on the GAL activity.[21] Recently, Gu and Zhou et al. utilized two-dimensional metal−organic framework nanosheets as effective enzyme inhibitors.[22]

Compared with NPs of large sizes (typically >10 nm), fluorescent ultrasmall nanoclusters (NCs) with core sizes of ∼2 nm, a representative class of ultrasmall nanoparticles (USNPs), were much less studied as nano regulators. However, they had some potential advantages. (i) NCs usually have a relatively small interface with enzymes like what the majority of biomolecular complexation events do, so that NCs may have a higher biological compatibility. In contrast, NPs have a larger interface with enzymes, therefore they may lead to a significant conformational change which denatures enzymes.[10,15] (ii) NCs have a larger curvature which may match well with the active sites or allosteric sites of some enzymes. (iii) NCs may be more effective due to their larger specific surface area. (iv) NCs have excellent fluorescence properties which can provide plenty of useful information to reflect the process of interactions.[30−32]

So, what happens when NCs meet enzymes? Will they affect enzyme activity? Do they share the same regulation mechanism with NPs? To address these issues, we selected gold nanoclusters (AuNCs) and bovine α-chymotrypsin (ChT) as the model system. AuNCs capped with dihydrolipoic acid (DHLA) and glutathione (GSH) respectively (DHLA-AuNCs and GSH-AuNCs) were chosen owing to their fluorescent properties, water solubility and stability. ChT is a digestive enzyme, belonging to the serine protease family which accounts for one-third of all known proteases. ChT has been widely used as the model enzyme with a clear structure and catalytic mechanism. It has totally 241 residues which are divided into three chains, with five disulfide bonds linking them as a whole. The active site of ChT contains a substrate recognition site and a catalytic site including the catalytic triad and the oxyanion hole. The catalytic triad consists of three key residues, including Ser195, Asp102 and His57. Among them, Asp102 forms a hydrogen bond with His57, enabling His57 a strong generalized base that deprotonates Ser195. Then, deprotonated Ser195 acts as the nucleophile to attack the carbonyl carbon of substrate's peptide bond to form a tetrahedral intermediate. The oxyanion hole, formed by Gly193 and Ser195, is positively charged to stabilize the oxyanion of the intermediate.[33,34] In this work, it was found that the surface modification of AuNCs played a critical role in the inhibition of ChT. DHLA-AuNCs inhibited the catalytic activity of ChT with high efficacy (>80%) and high potent (IC_{50} = 3.4 μM), whereas GSH-AuNCs had only little effect on the activity of ChT. Fluorescence spectroscopy, dynamic light scattering (DLS), agarose gel electrophoresis, inhibition kinetics experiment, and circular dichroism spectroscopy (CD) were performed to explore the inhibition mechanisms of DHLA-AuNCs. The results provided fundamental guidance for the design of nano inhibitors.

EXPERIMENTAL SECTION

Materials. Bovine α-chymotrypsin was purchased from Solarbio Life Sciences. The reduced L-glutathione (GSH), hydrogen tetrachloroaurate trihydrate ($HAuCl_4 \cdot 3H_2O$) and *N*-succinyl-L-phenylalanine-*p*-nitroanilide (SPNA) were purchased from Sigma-Aldrich. DL-α-lipoic acid (LA) was purchased from Tokyo Chemical Industry Co., Ltd. (TCI). A 0.01 M phosphate-buffered saline (PBS) solution prepared with deionized water (Millipore) was used in all experiments. Other chemicals were purchased from Sinopharm Chemical Reagent Co. (China). DHLA-AuNCs and GSH-AuNCs were synthesized according to the previous studies.[30−32,35,36]

Activity Assays. *N*-succinyl-L-phenylalanine-*p*-nitroanilide (SPNA) was used as the substrate of ChT (Supporting Information (SI) Scheme S1).[13−19] The hydrolysis product contains *p*-nitroaniline (PNA) whose characteristic absorption peak is around 380 nm. So that we used VICTOR X5 microplate reader (PerkinElmer) to monitor the increase of the absorbance at 405 nm (Abs405) to represent the reaction process. AuNCs (0−40 μM, 30 μL) and ChT (16 μM, 50 μL) were mixed at first. After incubation at 25 °C for 2 h, SPNA (1 mM, 120 μL) was added to the mixture. Abs405 was then read every 5 min for a total of 20 min. In order to explore the reversibility of the inhibition, NaCl (4 M, 0−10 μL) was added to the mixture of AuNCs (40 μM, 30 μL) and ChT (16 μM, 50 μL) before the 2 h incubation. After adding SPNA (1 mM, 120 μL), Abs405 was recorded every 5 min for a total of 20 min.

Fluorescence Spectroscopy. Three-dimensional fluorescence spectra of chymotrypsin were measured before and after the incubation with AuNCs. Fluorescence titrations were conducted at four different temperatures controlled by a circulating bath. Five μL ChT (2000 μM) was titrated into 2.5 mL AuNCs, followed with a sufficient stirring. Fluorescence spectra were measured after each titration. Synchronous fluorescence spectra of ChT were obtained with $\Delta\lambda$ = 15 or 60 nm since the beginning of incubation. All these experiments were performed with a Cary Eclipse fluorophotometer (Agilent). Thermally denatured chymotrypsin (DChT) obtained by heating in water bath at 60 °C for overnight was also measured as control.

Dynamic Light Scattering. Dynamic light scattering (DLS) measurements were carried out on a Nano-ZS90 zetasizer (Malvern) to determine the hydrodynamic sizes of ChT (8 μM) in the absence and presence of DHLA-AuNCs (8 μM) and GSH-AuNCs (8 μM). The set temperature was 298 K.

Fluorescence Lifetime. Fluorescence lifetime measurements of ChT (4 μM) were performed on a FLS1000 Photoluminescence Spectrometer (Edinburgh) before and after a 6 h incubation of AuNCs (4 μM). The fluorescence of ChT was excited by a laser (288 nm) and then collected in the range of 340−360 nm. DChT was also measured as control.

Agarose Gel Electrophoresis. Agarose gel electrophoresis was carried out with a MINI-PROTEAN Tetra System (BIO-RAD) to determine the binding ratio. Incubated samples with DHLA-AuNCs: ChT ratios ranging from 1:0 to 1:2 and GSH-AuNCs: ChT ratios from 1:0 to 1:10 at a constant concentration of AuNCs were electrophoresed at 80 V in the 1.5% gel (JY600C). The gel with DHLA-AuNCs was photographed by a camera and the photo was treated with black and white filter. Gel Doc XR+ Imaging System (BIO-RAD) was used to image the gel with GSH-AuNCs as GSH-AuNCs was not visible in natural light.

Inhibition Kinetics. Inhibition kinetics experiments were performed to judge the type of inhibition. After the incubation of AuNCs (0−40 μM, 30 μL) with ChT (16 μM, 50 μL), SPNA (0−1 mM, 120 μL) was added. Abs405 was then read every 6 min for a total of 24 min. DChT was also measured as control.

Circular Dichroism Spectroscopy. CD spectroscopy was performed on a circular dichroism photomultiplier (Applied Photophysics Limited, UK) to determine the secondary structure changes of ChT (6.4 μM) after incubation with AuNCs (25.6 μM) and Na_2SO_4 (0.13 M). DChT was also measured as control.

RESULTS AND DISCUSSION

The Inhibition of ChT Activity by DHLA-AuNCs. These two fluorescent nanoclusters herein have been well charac-

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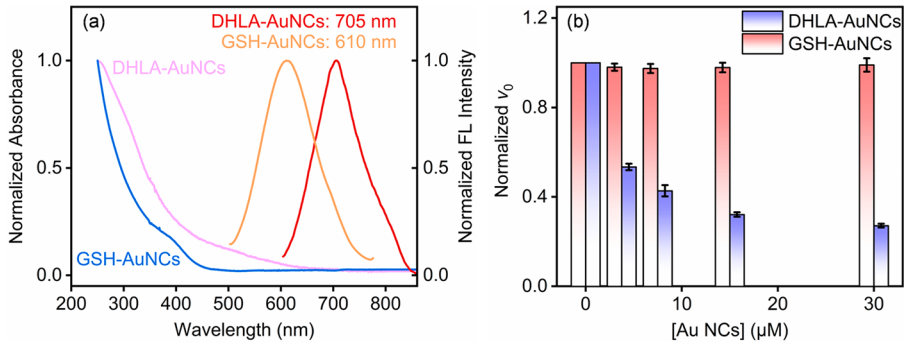


Figure 1. (a) Absorption and fluorescence emission (excitation at 565 nm for DHLA-AuNCs and 365 nm for GSH-AuNCs respectively) spectra of DHLA-Au NCs and GSH-AuNCs; (b) normalized rates (v_0) of ChT-catalyzed reactions under different concentrations of DHLA-AuNCs (blue) and GSH-AuNCs (red).

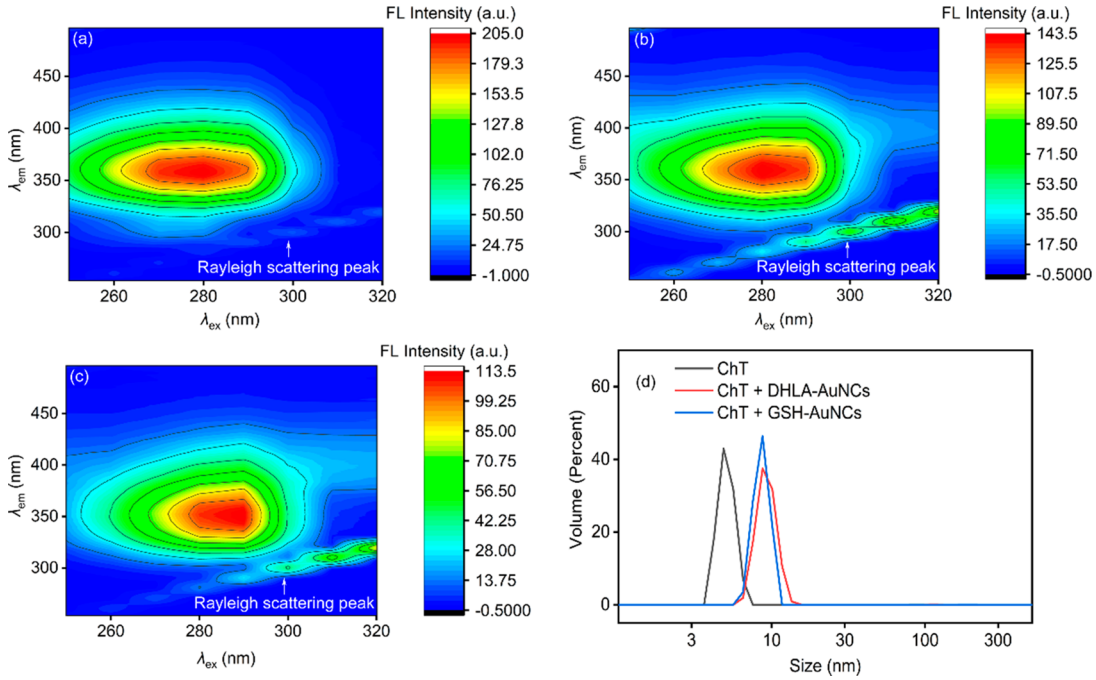


Figure 2. 3D fluorescence spectra and of ChT (a) and ChT incubated with DHLA-AuNCs (b) and GSH-AuNCs (c) and their hydrodynamic particle sizes (d).

terized in our previous studies.[30,32] Unlike the absorption spectra of AuNPs, both DHLA-AuNCs and GSH-AuNCs showed no surface plasmon resonance peak around 520 nm (Figure 1a), indicating that their sizes were below 2 nm. The emission maxima of DHLA-AuNCs and GSH-AuNCs were at ~705 nm and ~610 nm, respectively. ChT activity decreased significantly with the increase of DHLA-AuNCs (Figure 1b), indicating that DHLA-AuNCs had a high efficacy (~80%) to be a potent ChT inhibitor. In contrast, no significant inhibition of ChT was detected after incubation with GSH-AuNCs (Figure 1b) and both of the two ligands (SI Figure S1), namely DHLA and GSH (SI Scheme S2). These results suggested that surface ligands were critical for the interactions between enzymes and nanomaterials.[17] GSH-AuNCs had no inhibitory effect on ChT, presumably due to the zwitterionic nature of GSH. Likewise, as indicated by our previous studies, DHLA-AuNCs had stronger affinity toward human serum albumin (HSA) than GSH-AuNCs, owing to the hydrophobic interactions between DHLA-AuNCs and HSA.[30,32]

Binding of DHLA-AuNCs to ChT. Since binding was the most common interaction between enzymes and inhibitors, we assumed that binding was the predominant reason for inhibition. Three-dimensional fluorescence spectra were first measured to verify the binding of AuNCs to ChT. The Rayleigh scattering peak increased after the addition of AuNCs, indicating that ChT was bound to AuNCs to form a complex[23] (Figure 2a–c). DLS measurements were then carried out for further verification. The hydrodynamic size of ChT was ~5.2 nm, then it increased to ~9.3 nm and ~8.6 nm respectively after one equivalent DHLA-AuNCs or GSH-AuNCs was added, reflecting the possible formation of ChT-AuNCs complex (Figure 2d).[32] Besides, ChT might be dimerized as induced by AuNCs.[32]

A significant fluorescence quenching was observed when AuNCs were titrated to ChT (SI Figure S2). To study the quenching mechanism, the fluorescence lifetime of ChT was then measured before and after the incubation with AuNCs. Among the three fluorescent amino acid of ChT, Phe residues have the lowest absorptivity and fluorescence quantum yield

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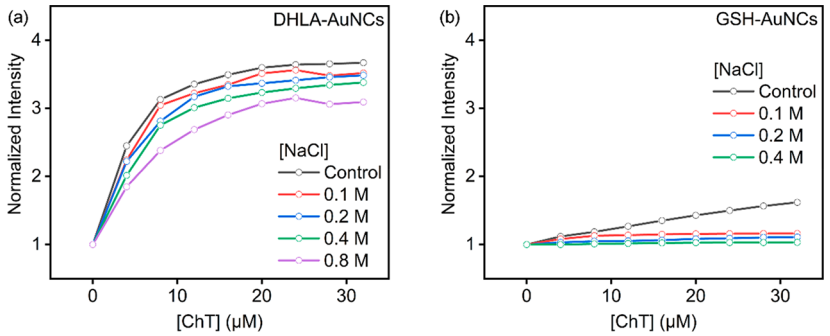


Figure 3. Fluorescence enhancement of DHLA-AuNCs (a) and GSH-AuNCs (b) with the titration of ChT under different concentrations of NaCl.

while the fluorescence of Tyr is often quenched due to its interactions with the peptide chain or the energy transfer to Trp. So, Trp with two different lifetimes was considered as the dominating fluorescence of ChT.[37] The fluorescence lifetime of ChT did not change after incubation with AuNCs, indicating a static quenching process when AuNCs were bound to ChT to form a ground state complex (SI Figure S3 and Table S1).[23] Nevertheless, whether it was the dominating reason for inhibition still remained unclear. In order to study the relationship between the binding and inhibition, fluorescence spectroscopic titration and enzyme activity assays were conducted in the presence of NaCl. Since the fluorescence of DHLA-AuNCs was enhanced with the titration of ChT, the fluorescence enhancement could reflect binding behaviors. The addition of NaCl led to a significant inhibition of the fluorescence enhancement of AuNCs, indicating that NaCl could weaken the binding between AuNCs and ChT (Figure 3). Notably, the addition of NaCl caused a more remarkable influence on the fluorescence enhancement of GSH-AuNCs. In the meantime, the addition of NaCl recovered the ChT activity, proving that binding between ChT and DHLA-AuNCs was the dominating reason for the inhibition of ChT (Figure 4). To be noted, NaCl alone could slightly accelerate the hydrolysis of SPNA, presumably because the increase of ionic strength could be beneficial for breaking chemical bonds.

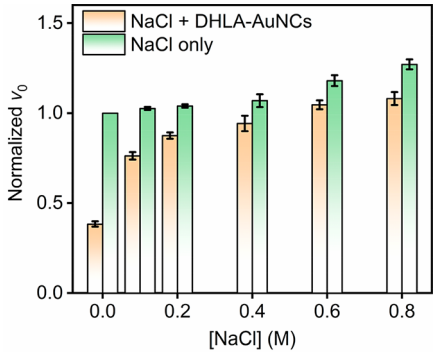


Figure 4. Normalized v_0 of ChT after being incubated with under different concentrations of NaCl.

The Binding Forces Between AuNCs and ChT. In order to establish the binding model of AuNCs and ChT, binding forces were studied. Figure 3 showed that the addition of NaCl only partially destroyed the binding between DHLA-AuNCs and ChT. In contrast, it inhibited the binding of GSH-AuNCs and ChT almost completely. As NaCl was able to destroy the electrostatic interactions by increased ionic strength, it could be inferred that DHLA-AuNCs were bound to ChT through electrostatic forces and other interaction forces, while electrostatic forces were the dominating forces between GSH-AuNCs and ChT.

This deduction matches well with the structure (SI Scheme S2) of the two ligands. In PBS (pH 7.4), ChT is positively charged with its isoelectric point around 8.2.[38] DHLA is negatively charged with its pK_a approximately equal to 5.4. It has one negatively charged carboxyl which can contribute to electrostatic forces and a hydrophobic chain which may have hydrophobic interactions with ChT.[39] As for GSH, it is also negatively charged with its isoelectric point around 6.0 and has 2 carboxyl groups which are prone to bind ChT through electrostatic forces.[40]

Fluorescence spectroscopic titrations were then performed under four different temperatures to obtain the thermodynamic parameters of binding processes. The fluorescence intensity of AuNCs increased upon the titration of ChT (SI Figure S4). Hill equation was then used to describe the relationship between the concentration of ChT and the fluorescence intensity (SI Figure S5).[41]

$$F = F_0 + \frac{F_{max} - F_0}{1 + (K_d/[ChT])^n} \quad (1)$$

Here, F and F_0 are the fluorescence intensities of AuNCs in the presence and absence of ChT respectively. F_{max} is the maximum fluorescence intensity of AuNCs in the presence of ChT. K_d is the dissociation constant that quantifies the strength of the ChT-AuNCs interaction. n is the Hill coefficient that describes the cooperativity of association. Our data showed a great match in the nonlinear curve fitting with Hill equation. The K_d values at different temperatures were thus obtained. The van't Hoff equation describes the relationship between equilibrium constant and thermodynamic parameters in the following formula:[30]

$$\ln K_a = -\frac{\Delta H}{RT} + \frac{\Delta S}{R} \quad (2)$$

Here, K_a refers to the association constant, R is the gas constant, T is the absolute temperature, and ΔH and ΔS stand for the enthalpy and entropy change, respectively. This equation was then used to obtain the thermodynamic parameters (Figure 5) with the results summarized in Table 1. The Hill coefficients for all the AuNCs-ChT systems were approximately equal to 1, indicating that the binding of AuNCs with ChT had little effect on the affinity of AuNCs for ChT. The association constant of DHLA-AuNCs with ChT was ~1 × 10^5 L·mol^{-1} (298 K), indicating that DHLA-AuNCs had a

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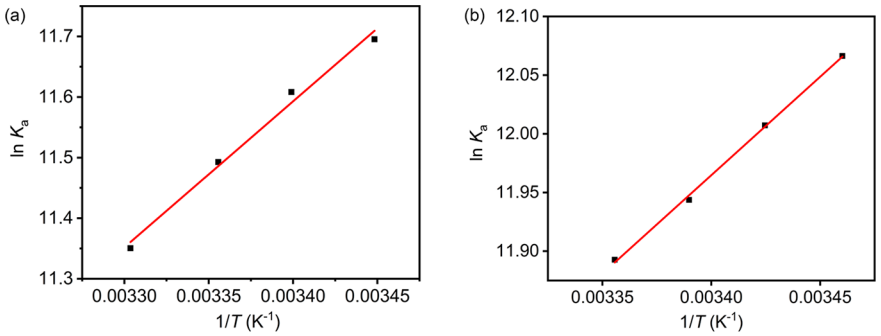


Figure 5. Van't Hoff equation of the association constant for DHLA-AuNCs (a) and GSH-AuNCs (b) as a function of the reciprocal of temperature.

Table 1. Thermodynamic Parameters of the Association of ChT and AuNCs

AuNCs	T (K)	n	K_a (10^5 L mol^{-1})	R^2	ΔH (kJ mol^{-1})	ΔS (J mol^{-1}·K^{-1})	ΔG (kJ mol^{-1})	R^2
DHLA-AuNCs	290	1.14	1.20	0.997	−20.0	28.3	−28.2	0.986
	294	1.06	1.10	0.998			−28.4	
	298	0.99	0.98	0.997			−28.5	
	303	1.11	0.85	0.998			−28.6	
GSH- AuNCs	289	1.29	1.74	0.995	−13.9	52.0	−29.0	0.998
	292	1.40	1.64	0.994			−29.1	
	295	1.41	1.54	0.995			−29.3	
	298	1.30	1.46	0.993			−29.5	

relatively strong binding with ChT, similar to the result that DHLA-QDs were easy to form protein corona.[42] According to Ross theory,[43] the negative ΔH and positive ΔS indicated that the main interaction forces between both the two AuNCs and ChT were electrostatic forces.

Agarose gel electrophoresis (AGE) was carried out to determine the binding ratios. In Figure 6a, only DHLA-AuNCs were visible. As the ChT: DHLA-AuNCs ratio increased from 0:1 to 0.6:1, a significant lag occurred in the DHLA-AuNCs bands, indicating a decrease of free DHLA-AuNCs and the possible formation of complex composed by DHLA-AuNCs and ChT. As the ChT: DHLA-AuNCs ratio continued to increase from 0.8:1 to 2:1, DHLA-AuNCs remained almost the same electrophoresis rate, indicating that the binding ratio of ChT and DHLA-AuNCs was ∼ 1:1. In Figure 6b, only GSH-AuNCs were visible. As the ChT: GSH-AuNCs ratio increased to 2:1, the band of complex composed of ChT and GSH-AuNCs appeared. Free GSH-AuNCs were still found even though the ChT: GSH-AuNCs ratio reached ∼10:1, maybe due to the weak binding between GSH-AuNCs and ChT. Therefore, the binding ratio of ChT to GSH-AuNCs was not available by AGE. However, considering that GSH-AuNCs had a similar structure with DHLA-AuNCs, we hypothesized that the binding ratio of ChT to both AuNCs was ∼ 1:1.

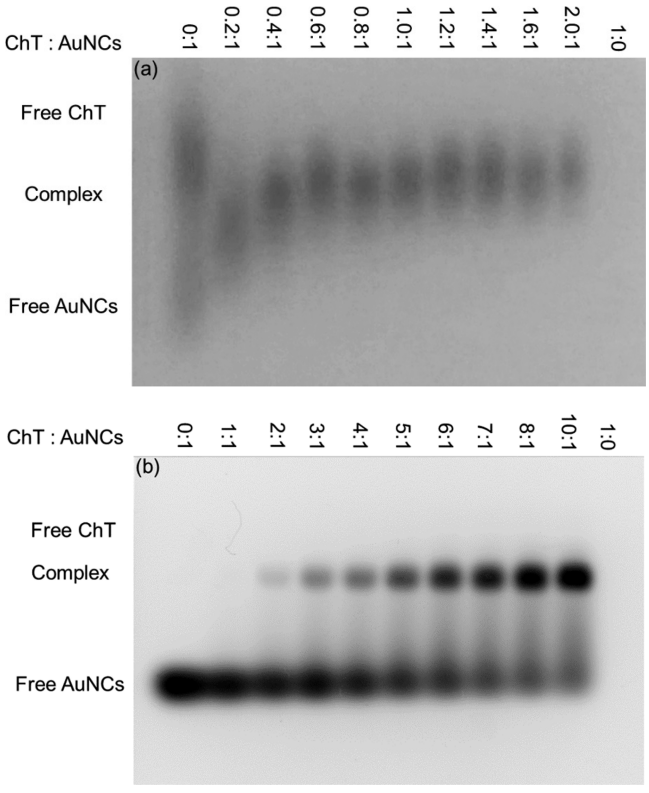


Figure 6. Agarose gel electrophoresis of DHLA-AuNCs (a) and GSH-AuNCs (b) under different concentrations of ChT. Free ChT was not visible in this experiment.

The Binding Site of AuNCs on ChT. ChT structure (PDB ID: 1OXG[44]) and its surface potential calculated by APBS[45] with the help of PDB2PQR[46] were displayed in VMD (http://www.ks.uiuc.edu). ChT has four tyrosine (Tyr) residues and eight tryptophan (Trp) residues (Figure 7a). Tyr94, Tyr146, Tyr171, and Trp172 are close to the active site. Trp215 and Tyr228 are residues of the substrate binding site deeply inside ChT molecule, and the other five Trp residues are distributed away from the active site (Figure 7b). In general, the surface potential of ChT is distributed like a coin. The positive charge is concentrated around the triangle with Tyr94, Tyr146, and Tyr171 as its vertexes (Figure 7c), while the other side is generally negatively charged (Figure 7d).

Synchronous fluorescence spectra were measured to explore the relationship between the binding site and the above

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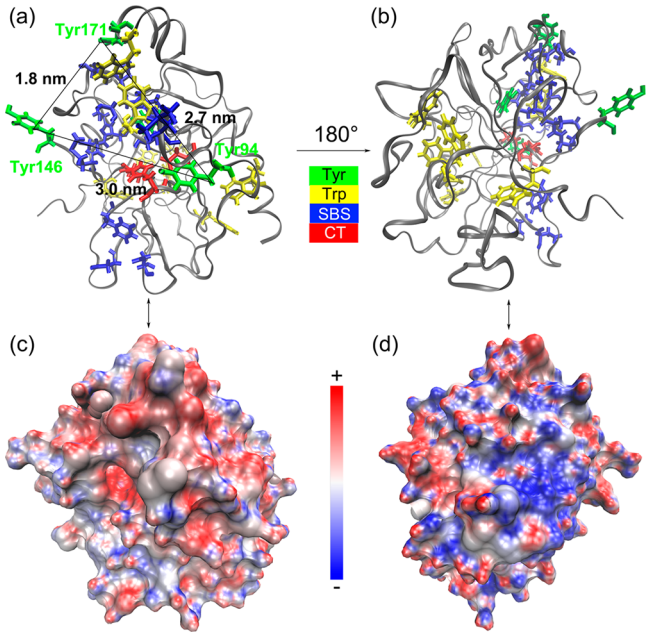


Figure 7. Schematic representations of ChT before (a) and after (b) rotating 180°. Surface potential diagrams of ChT before (c) and after (d) rotating 180°. The catalytic triad (CT), substrate binding site (SBS), Trp residues, and Tyr residues are shown in red, blue, yellow, and green, respectively. This image was made with (Visual Molecular Dynamics) VMD. VMD is developed with NIH support by the Theoretical and Computational Biophysics group at the Beckman Institute, University of Illinois at Urbana−Champaign.

residues (Figure 8). Herein, when $\Delta\lambda$ was set as 15 nm, it reflected the fluorescence of Tyr. When $\Delta\lambda$ was set as 60 nm, it

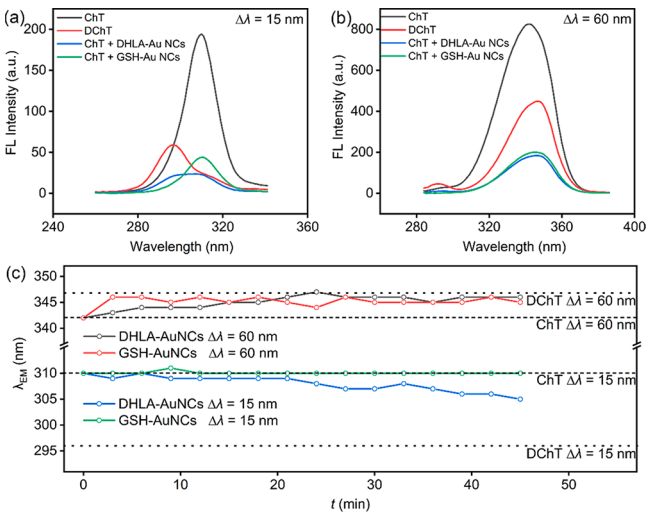


Figure 8. Synchronous fluorescence spectra of ChT titrated with AuNCs with $\Delta\lambda$ = 15 nm (a) and 60 nm (b); the changes in the emission maxima of ChT in the presence of AuNCs over time (c).

reflected the fluorescence of Trp. DHLA-AuNCs caused a significant blueshift of the emission peak of Tyr while GSH-AuNCs had little effect, indicating that DHLA-AuNCs caused a more significant decrease in the polarity of the microenvironment of Tyr (Figure 8a). The possible reason was that Tyr residues were bound to DHLA-AuNCs through hydrophobic interactions. DHLA-AuNCs and GSH-AuNCs both led to a significant redshift to the emission peak of Trp, indicating an increase in the polarity of the microenvironment of Trp (Figure 8b). This could be explained as that Trp residues were bound to the polar regions of AuNCs through electrostatic forces. The blueshift rate of Tyr caused by DHLA-AuNCs was slower than the redshift rate of Trp, indicating that they corresponded to two different processes (Figure 8c). According to the above results, we inferred these two steps as that Trp was bound to the polar region of DHLA-AuNCs at first, followed by Tyr's binding to the hydrophobic region of DHLA-AuNCs.

As both DHLA-AuNCs and GSH-AuNCs were first bound to ChT through electrostatic forces, we inferred that both AuNCs were bound onto the positively charged areas of ChT, which was highly overlapped with the triangular region consisting of Tyr94, Tyr146, and Tyr171 as the vertexes. The reasons were as follows. (i) The side lengths of this triangular area were 1.8 nm, 2.7 nm, and 3.0 nm, respectively, matching well with the size of AuNCs (Figure 7a). (ii) The triangular area was the region where positive charge was most concentrated on ChT molecule, which contributed to the electrostatic forces since DHLA-AuNCs were negatively charged in PBS (pH 7.4) (Figure 7c). (iii) The triangular area contained three Tyr residues and two Trp residues, consistent with the results of synchronous fluorescence spectra (Figure 8). (iv) This triangular area was adjacent to the substrate binding site of ChT, which was important to enzyme catalysis (Figure 7a).

Noncompetitive Inhibition of ChT Activity by DHLA-AuNCs. The addition of NaCl recovered ChT activity almost completely after it was inhibited by DHLA-AuNCs, indicating DHLA-AuNCs inhibited ChT reversibly. There are three fundamental types of reversible inhibition, including competitive inhibition, uncompetitive inhibition, and noncompetitive inhibition.[6] To study the inhibition type of ChT by DHLA-AuNCs, inhibition kinetics experiments were carried out. In enzymology, Michaelis−Menten equation is one of the best-known equations to describe the relationship between reaction rate and the concentration of substrate.[6] The equation is given as follows:

$$v = \frac{\mathrm{d}[\mathrm{P}]}{\mathrm{d}t} = \frac{v_{\max}[\mathrm{S}]}{K_{\mathrm{M}} + [\mathrm{S}]} \tag{3}$$

Herein, v represents the reaction rate. [P] and [S] represent the concentration of product and substrate, respectively. $v_{\max}$ represents the maximum rate at saturation concentration of the substrate. The Michaelis constant K_{M} is the substrate concentration at which the reaction rate is half of $v_{\max}$. We can also infer that K_{M} can describe the affinity of an enzyme toward a substrate, that is, the larger K_{M} is, the lower the affinity is. Michaelis−Menten model is also a useful tool to describe inhibition kinetics and the equation used here is modified as follows:

$$v = \frac{v_{\max}^{\mathrm{app}}[\mathrm{S}]}{K_{\mathrm{M}}^{\mathrm{app}} + [\mathrm{S}]} \tag{4}$$

Herein, $v_{\max}^{\mathrm{app}}$ and $K_{\mathrm{M}}^{\mathrm{app}}$ represent the apparent maximum rate and the apparent Michaelis constant, respectively. In competitive inhibition, an inhibitor bind to the active site of the enzyme, competitively hindering the binding of a substrate and the enzyme. As a result, $v_{\max}^{\mathrm{app}}$ remains the same because the inhibition caused by the competitive inhibitor can be

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conquered by sufficient substrate. However, K_M^{app} increases because more substrate is necessary to reach v_{max}^{app}. As for uncompetitive inhibition, the inhibitors bind only to the enzyme−substrate complex, blocking the process of product generation. Therefore, v_{max}^{app} decreases with the decrease of activated ES complex. K_M^{app} also decreases due to the higher binding affinity of ES complex. In noncompetitive inhibition, the binding of an inhibitor to the enzyme reduces the enzymatic activity but has no effect on the binding of substrate to the enzyme. Thus, v_{max}^{app} decreases while K_M remains unaltered. Lineweaver−Burk (LB) equation is the double reciprocal form of Michaelis−Menten equation.

$$\frac{1}{v} = \frac{K_M^{app}}{v_{max}^{app}} \times \frac{1}{[S]} + \frac{1}{v_{max}^{app}} \quad (5)$$

We fitted our data with LB equation to calculate the values of K_M^{app} and v_{max}^{app} at different concentration of inhibitors (Figure 9). The extended lines all intersected at the same point on the

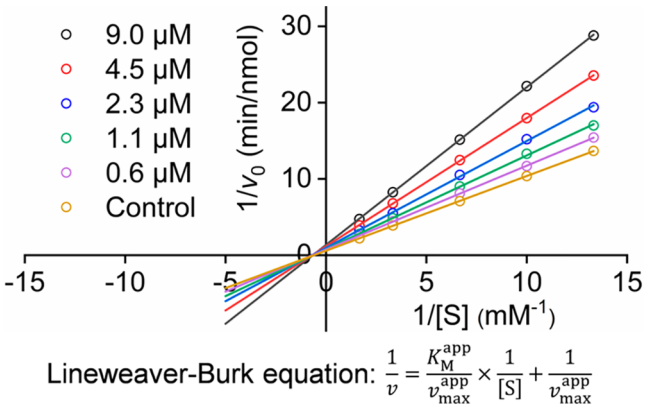


Figure 9. Lineweaver−Burk plots of ChT under different concentrations of DHLA-AuNCs.

X-axis, indicating that DHLA-AuNCs inhibited ChT activity with a noncompetitive mechanism. The results summarized in Table 2 showed that v_{max}^{app} decreased with the increase of

Table 2. Apparent Maximum Rate and the Apparent Michaelis Constant under Different Concentrations of DHLA-AuNCs

[AuNCs] (μM)	0	0.4	0.8	1.5	3.0	6.0
K_M^{app} (mM)	1.69	1.51	1.58	1.45	1.45	1.54
v_{max}^{app} (nmol/min)	1.72	1.37	1.29	1.03	0.86	0.74
R^2	0.999	0.999	0.999	0.998	0.999	0.999

inhibitors while K_M^{app} remained almost constant, verifying that the noncompetitive inhibition model could be applied to describe the inhibition of ChT activity caused by DHLA-AuNCs.

Fluorescence titration was then carried out to further support the noncompetitive inhibition mechanism (SI Figure S6). With the titration by ChT, the fluorescence intensity of DHLA-AuNCs significantly increased due to their interactions. No changes were found with the addition of SPNA before and after the titration, indicating that SPNA did not affect the interactions between ChT and DHLA-AuNCs. This finding was consistent with the noncompetitive inhibition mechanism in which inhibitors did not bind onto the active site of ChT.

Another interesting phenomenon worth mentioning was that DHLA-AuNCs only partially inhibited ChT activity (Figure 3). As the concentration of DHLA-AuNCs increased, the inhibition ratio reached up to an upper limit of 80%. In competitive inhibition, however, inhibitors were bound to the active site of enzymes, blocking the binding of enzymes with substrates. As a result, the enzymatic activity could be completely inhibited with sufficient competitive inhibitors.[26−29] Therefore, the partial inhibition also became powerful evidence for noncompetitive inhibition.

This noncompetitive inhibition mechanism meant that DHLA-AuNCs were bound onto ChT at a site rather than the active site so that the binding of DHLA-AuNCs and SPNA to ChT would not affect each other. Figure 7 showed that the positively charged triangular region was highly overlapped with the substrate binding site. SPNA, a tripeptide analogue, was so small that it occupied a very small portion of the substrate binding site. Besides, it was critical that AuNCs had very small sizes (~2 nm). As a result, the binding of DHLA-AuNCs and SPNA to ChT hardly affected each other.

For noncompetitive inhibitions, the maximum rate and its apparent value obey the following equation:

$$v_{max}^{app} = \frac{v_{max}}{1 + \frac{[I]}{K_i}} \quad (6)$$

Herein, K_i represents the inhibition constant that quantitatively describes the binding affinity between inhibitors and enzymes or enzyme−substrate complexes. eq 6 can be transformed as follows.

$$\frac{1}{v_{max}^{app}} = \frac{[I]}{v_{max}K_i} + \frac{1}{v_{max}} \quad (7)$$

Therefore, we took the concentration of DHLA-AuNCs as the *X*-axis and the reciprocals of the apparent maximum rate as the *Y*-axis (SI Figure S7). So, the inhibition constant was equal to the value of intercept over slope. As the result showed, the inhibition constant of DHLA-AuNCs to ChT was 3.4 μM. In noncompetitive inhibition, the inhibitory potency (IC_{50}) was equal to K_i with a value of 3.4 μM, demonstrating a relatively high potency.

Among all categories of ChT inhibitors, chymostatin, a peptide-derived aldehyde, has the highest potency with K_i of 0.4 nM. Proteins such as Bowman-Birk inhibitors and wild emmer chymotrypsin inhibitors have inhibition constants at nM level. Small molecules like arylboronic acids, analogues of phenylalanine ethyl eater, and phenylarsonate have a wide range of K_i from nM to mM level.[47−52]

Unlike competitive inhibitors, noncompetitive inhibitors can always retain partial activity, reducing the risk of being largely taken by mistake. Besides, noncompetitive inhibitors are not affected by the substrate concentration, so that they can be more effective than competitive inhibitors under high concentrations of substrate. However, noncompetitive inhibitors usually act on the allosteric sites which are less conserved than the active sites, so that noncompetitive inhibitors may have different effects on the same enzymes of different species.

Combining our work and previous researches, we concluded that the majority of nano inhibitors inhibited enzyme activity through competitive inhibition mechanism. To design other types of inhibitors, smaller interface was needed in order not to affect the substrate binding sites. For example, *p*-mercaptobenzoic acid-coated AuNPs with core size of 2.0 ~ 2.5 nm

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inhibited thrombin through a mixed-type inhibition mechanism with uncompetitive inhibition being predominant.[53] Furthermore, carbon nanotubes with low surface modification density exhibited noncompetitive characteristics while those with high surface modification density competitively inhibited ChT.[20]

Allosteric Inhibition Responsible for the Noncompetitive Mechanism. What was the exact reason behind this noncompetitive inhibition mechanism? Why did inhibition occur after DHLA-AuNCs bind to the positively charged triangular region without affecting substrate binding? Noncompetitive inhibitors inhibited enzyme activity mainly through structural change of the catalytic site and steric hindrance that prevented the release of products. However, as both DHLA-AuNCs and GSH-AuNCs were bound to almost the same site of ChT, steric hindrance seemed unlikely to account for the inhibitory effects. In this regard, CD spectroscopy was then employed to detect the structural changes of ChT caused by AuNCs (Figure 10). ChT had two

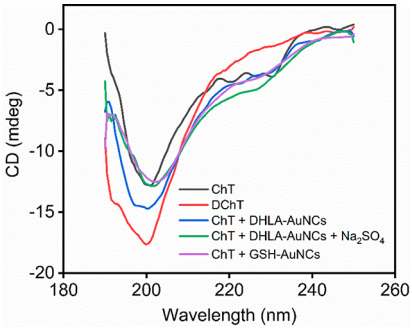


Figure 10. CD spectra of ChT before and after incubation with AuNCs.

characteristic minima at 201 and 232 nm, while denatured ChT (DChT) had only one minimum at 198 nm. The addition of GSH-AuNCs had no effect on the minimum at 201 nm but caused a slight decrease to the minimum at 232 nm, indicating that there was only a slight change in the secondary structure of ChT. In distinct contrast, the addition of DHLA-AuNCs partially decreased the minimum at 232 nm and caused a remarkable change in the one at 201 nm, indicating a significant change in the secondary structure of ChT.

Since NaCl had a strong signal in the range of 190 ~ 200 nm in the CD spectra, while Na_2SO_4 had no interference in this range (SI Figure S8), NaCl was replaced by Na_2SO_4 to increase the ionic strength in this experiment. Activity assays proved that Na_2SO_4 had the same effect as NaCl to destroy the binding between ChT and DHLA-AuNCs under the same ionic strength with NaCl and Na_2SO_4 (SI Figure S9). The addition of Na_2SO_4 eliminated almost all the changes in the spectrum, indicating that Na_2SO_4 destroyed the electrostatic interactions between DHLA-AuNCs and ChT, recovering the secondary structure of ChT changed by DHLA-AuNCs.

Fluorescence spectroscopic titration can also provide a powerful proof for the structural changes of ChT. As SI Figure S2a shows, the intrinsic fluorescence of ChT was quenched significantly by the titration of AuNCs, and the characteristic fluorescence emission peak of natural ChT at 338 nm exhibited a redshift, indicating a structural change. Then, kinetics experiments were performed to study the structural change process of ChT in the presence of AuNCs. Emission maximum λ_m of ChT was measured after each titration. As thermally denatured ChT (DChT) had its characteristic fluorescence emission peak at 360 nm, there existed a 22 nm redshift between natural ChT and its denatured form. Thus, the structural change fraction ΔZ could be calculated by the following formula:

$$\Delta Z = \frac{\lambda_m - 338}{22} \tag{8}$$

The results showed that the rate of structural change of ChT was highly dependent on the temperature, concentration and types of AuNCs (Figure 11 and SI Figure S10). Both the

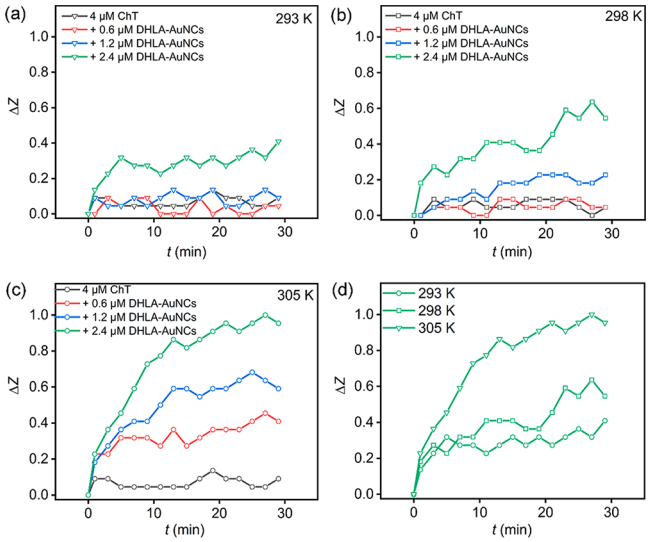


Figure 11. Structural change rate of ChT under different concentrations of DHLA-AuNCs at 293 K (a), 298 K (b), and 305 K (c). Structural changes of ChT treated with 2.4 μM DHLA-AuNCs at different temperatures (d).

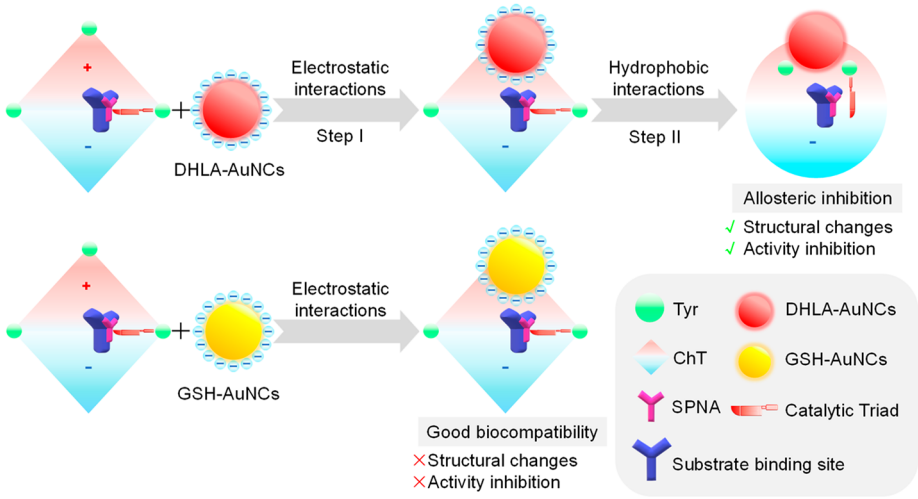
higher temperature and the larger concentration of DHLA-AuNCs could result in faster structural changes of ChT (Figure 11). DHLA-AuNCs could significantly change the structure of ChT. In contrast, GSH-AuNCs only caused very minor effect on it (SI Figure S10). These results were well consistent with the CD spectroscopic studies.

Our recent study showed that AuNCs as a representative class of USNPs might form protein complex in the biological milieu.[32] These complexes were also found in the AGE experiments of this work (Figure 6). Allosteric effect caused by nanobio complex was one of the reasonable assumptions. DHLA-AuNCs were first bound to the positively charged triangular region, considered as the allosteric site of ChT, through electrostatic forces, then the three Tyr residues at the vertexes of that triangular region moved close to the hydrophobic chain of DHLA driven by hydrophobic forces, leading to the structural change and thereby the inhibition of ChT activity. Apart from allosteric effect, ligand exchange may also explain this phenomenon. However, previous study indicated that ChT became completely unfolded after being used as the stabilizer of AuNCs,[54] which differed from our experimental results. Therefore, allosteric effect became the most possible reason for ChT inhibition by DHLA-AuNCs. On the contrary, GSH-AuNCs had no such hydrophobic interactions with ChT due to the zwitterionic nature of GSH. In principle, this demonstrated the better biocompatibility of

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Scheme 1. Two-Step Model for the Interactions of ChT with DHLA-AuNCs[a]



[a]Step I: Electrostatic interactions; Step II: Hydrophobic interactions. GSH-AuNCs had no hydrophobic interactions with ChT due to the zwitterionic nature of GSH.

fluorescent GSH-AuNCs when applied in biomedical fields. The mechanism is concluded as a two-step interaction model (Scheme 1).

To the best of our knowledge, our work provides the first example of allosteric inhibition of ChT by nano regulators. Previous literature has reported that dihydroindolizine (DHI) allosterically inhibited ChT based on the results of molecular docking simulations.[55] The allosteric inhibition mechanism of some other enzymes belonging to the serine protease family has been reported as well.[56−59] Analogues of sulfated nonsaccharide glycosaminoglycan mimetics allosterically inhibited human plasmin. Moreover, *p*-mercaptobenzoic acid-coated ultrasmall gold NPs, analogues of sulfated dehydropolymers of 4-hydroxycinnamic acids, sulfated coumarins and sulfated benzofurans allosterically inhibited thrombin activity. The allosteric regulation of thrombin was widely studied with its two exosites being identified already,[53] so that it was reasonable for ChT to be allosterically inhibited by DHLA-AuNCs since ChT had a similar structure with thrombin.

In order to develop nano regulators, more efforts should be devoted to the accurate design of NCs according to the structures and properties of active sites and allosteric sites. As it is difficult to reveal the interaction mechanisms between enzymes and their regulators with the existing experimental methods, molecular simulations will exhibit their importance in future studies.[60−64]

CONCLUSIONS

In summary, the interactions of DHLA- and GSH-AuNCs with chymotrypsin were investigated with thermodynamic and kinetic perspectives. Kinetics studies showed that DHLA-AuNCs can inhibit more than 80% of ChT activity through noncompetitive inhibition, whereas GSH-AuNCs had no significant effect, indicating that surface modifications of nanomaterials played critical roles in the regulation of ChT activity. The AuNCs and ChT were bound together by electrostatic forces to form possible protein complexes with a stoichiometry of ∼1:1 (AuNCs/ChT). The positively charged triangular region of ChT with three Tyr residues at the vertexes might be the binding site. CD and fluorescence spectroscopic studies demonstrated significant structural changes of ChT caused by DHLA-AuNCs. A two-step mechanism was proposed as electrostatic interactions at first, followed by hydrophobic interactions between ChT and DHLA-AuNCs. This work underlined the importance of hydrophobic forces with nano inhibitors, which should be instructive for the design of nano regulators. Besides, it demonstrated the good biocompatibility of GSH-AuNCs in a perspective of nano-safety.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.langmuir.0c00699.

Hydrolytic process of SPNA catalyzed by ChT (Scheme S1); Chemical structures of DHLA and GSH (Scheme S2); Influence of ligands on ChT activity (Figure S1); Fluorescence quenching of ChT titrated with AuNCs (Figure S2); Fluorescence decay traces of DChT and ChT in the absence or presence of Au NCs (Figure S3); Fluorescence lifetimes of ChT before and after being treated with AuNCs (Table S1); Fluorescence spectra of DHLA-Au NCs titrated by ChT at different temperatures (Figure S4); Nonlinear curve fitting of AuNCs fluorescence intensity with Hill equation (Figure S5); Fluorescence enhancement of DHLA-AuNCs titrated with ChT in the presence of SPNA (Figure S6); Linear relationship between DHLA-AuNCs concentrations and the reciprocals of the apparent maximum rate (Figure S7); CD spectra of 10 different salts (Figure S8); Normalized v_0 of ChT in the absence and presence of DHLA-AuNCs, NaCl and Na_2SO_4 (Figure S9); Structural change rate of ChT under different concentrations of GSH-AuNCs at different temperatures (Figure S10) (PDF)

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) Denton, R. M. Regulation of Mitochondrial Dehydrogenases by Calcium Ions. *Biochim. Biophys. Acta, Bioenerg.* **2009**, *11*, 1309−1316.

(2) Ribatti, D. Endogenous Inhibitors of Angiogenesis: A Historical Review. *Leuk. Res.* **2009**, *33*, 638−644.

(3) Stark, A.; Sriskantharajah, S.; Hessel, E. M.; Okkenhaug, K. PI3K Inhibitors in Inflammation, Autoimmunity and Cancer. *Curr. Opin. Pharmacol.* **2015**, *23*, 82−91.

(4) Robert, R. Janus Kinase (JAK) Inhibitors in the Treatment of Inflammatory and Neoplastic Diseases. *Pharmacol. Res.* **2016**, *111*, 784−803.

(5) Nussinov, R.; Tsai, C. Allostery in Disease and in Drug Discovery. *Cell* **2013**, *153*, 293−305.

(6) Copeland, R.; Enzymes, A. *A Practical Introduction to Structure, Mechanism, and Data Analysis*; Wiley-VCH: New York, 2000.

(7) Lisi, G. P.; Loria, J. P. Allostery in Enzyme Catalysis. *Curr. Opin. Struct. Biol.* **2017**, *47*, 123−130.

(8) Schwab, A.; Illarionov, B.; Frank, A.; Kunfermann, A.; Seet, M.; Bacher, A.; Witschel, M. C.; Fischer, M.; Groll, M.; Diederich, F. Mechanism of Allosteric Inhibition of the Enzyme IspD by Three Different Classes of Ligands. *ACS Chem. Biol.* **2017**, *12*, 2132−2138.

(9) Changeux, J. P. 50 Years of Allosteric Interactions: The Twists and Turns of the Models. *Nat. Rev. Mol. Cell Biol.* **2013**, *14*, 819−829.

(10) Chen, M.; Zeng, G.; Xu, P.; Lai, C.; Tang, L. How Do Enzymes 'Meet' Nanoparticles and Nanomaterials? *Trends Biochem. Sci.* **2017**, *42*, 914−930.

(11) Kaplan, H.; Laidler, K. J. Kinetics of α-Chymotrypsin Action. III. Mechanisms of Inhibition. *Can. J. Chem.* **1967**, *45*, 559−565.

(12) Li, D.; Zhang, W.; Yu, X.; Wang, Z.; Su, Z.; Wei, G. When Biomolecules Meet Graphene: From Molecular Level Interactions to Material Design and Applications. *Nanoscale* **2016**, *8*, 19491−19509.

(13) Fischer, N. O.; Mcintosh, C. M.; Simard, J. M.; Rotello, V. M. Inhibition of Chymotrypsin through Surface Binding Using Nanoparticle-based Receptors. *Proc. Natl. Acad. Sci. U. S. A.* **2002**, *99*, 5018−5023.

(14) Fischer, N. O.; Verma, A.; Goodman, C. M.; Simard, J. M.; Rotello, V. M. Reversible "Irreversible" Inhibition of Chymotrypsin Using Nanoparticle Receptors. *J. Am. Chem. Soc.* **2003**, *125*, 13387−13391.

(15) Hong, R.; Nicholas, O. F.; Ayush, V. Control of Protein Structure and Function through Surface Recognition by Tailored Nanoparticle Scaffolds. *J. Am. Chem. Soc.* **2004**, *126*, 739−743.

(16) Hong, R.; Emrick, T.; Rotello, V. M. Monolayer-controlled Substrate Selectivity Using Noncovalent Enzyme-nanoparticle Conjugates. *J. Am. Chem. Soc.* **2004**, *126*, 13572−13573.

(17) You, C. C.; De, M.; Han, G.; Rotello, V. M. Tunable Inhibition and Denaturation of α-Chymotrypsin with Amino Acid-functionalized Gold Nanoparticles. *J. Am. Chem. Soc.* **2005**, *127*, 12873−12881.

(18) You, C. C.; Agasti, S. S.; De, M.; Knapp, M. J.; Rotello, V. M. Modulation of the Catalytic Behavior of α-Chymotrypsin at Monolayer-protected Nanoparticle Surfaces. *J. Am. Chem. Soc.* **2006**, *128*, 14612−14618.

(19) De, M.; Chou, S.; Dravid, V. P. Graphene Oxide as an Enzyme Inhibitor: Modulation of Activity of α-Chymotrypsin. *J. Am. Chem. Soc.* **2011**, *133*, 17524−17527.

(20) Zhao, X.; Hao, F.; Liu, D. Influence of the Surface Functional Group Density on the Carbon-nanotube-induced α-Chymotrypsin Structure and Activity Alterations. *ACS Appl. Mater. Interfaces* **2015**, *7*, 18880−18890.

(21) Cha, S. H.; Hong, J.; McGuffie, M.; Yeom, B.; VanEpps, J. S.; Kotov, N. A. Shape-dependent Biomimetic Inhibition of Enzyme by Nanoparticles and Their Antibacterial Activity. *ACS Nano* **2015**, *9*, 9097−9105.

(22) Xu, M.; Yuan, S.; Chen, X. Y.; Chang, Y. J.; Day, G.; Gu, Z. Y.; Zhou, H. C. Two-dimensional Metal-organic Framework Nanosheets as an Enzyme Inhibitor: Modulation of the α-Chymotrypsin Activity. *J. Am. Chem. Soc.* **2017**, *139*, 8312−8319.

(23) Ellappan, V.; Kesavan, M.; Ramalingam, P.; Kulandaivel, J.; Rajalingam, R. Interaction of Digestive Enzymes with Tunable Light Emitting Quantum Dots: A Thorough Spectroscopic Investigation. *Luminescence* **2015**, *30*, 978−989.

(24) Peng, J.; Liu, S.; Yan, S.; Fan, X.; He, Y. A Study on the Interaction between CdTe Quantum Dots and Chymotrypsin Using Optical Spectroscopy. *Colloids Surf., A* **2010**, *359*, 13−17.

(25) Shahnaz, A.; Tahereh, T. M. Interaction of Silver Nanoparticles with Lysozyme: Functional and Structural Investigations. *Surf. Interfaces* **2018**, *10*, 216−221.

(26) Koushik, S.; Parthasarathi, D. Exfoliated Nanosheets of a Cu^{II} Coordination Polymer Modulate Enzyme Activity of α-Chymotrypsin. *Chem. - Eur. J.* **2018**, *24*, 1−7.

(27) Li, H.; Kong, W.; Liu, J.; Yang, M.; Huang, H.; Liu, Y.; Kang, Z. Carbon Dots for Photoswitching Enzyme Catalytic Activity. *J. Mater. Chem. B* **2014**, *2*, 5652−5658.

(28) Tabish, T. A.; Pranjol, M. Z. I.; Karadag, I.; Horsell, D. W.; Whatmore, J. L.; Zhang, S. Influence of Luminescent Graphene Quantum Dots on Trypsin Activity. *Int. J. Nanomed.* **2018**, *13*, 1525−1538.

(29) Zhang, B.; Xing, Y.; Li, Z.; Zhou, H.; Mu, Q.; Yan, B. Functionalized Carbon Nanotubes Specifically Bind to α-Chymotrypsin's Catalytic Site and Regulate Its Enzymatic Function. *Nano Lett.* **2009**, *9*, 2280−2284.

(30) Yin, M. M.; Dong, P.; Chen, W. Q.; Xu, S. P.; Yang, L. Y.; Jiang, F. L.; Liu, Y. Thermodynamics and Mechanisms of the Interactions between Ultrasmall Fluorescent Gold Nanoclusters and Human

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Serum Albumin, γ-Globulins, and Transferrin: A Spectroscopic Approach. *Langmuir* **2017**, *33*, 5108−5116.
(31) Yin, M. M.; Chen, W. Q.; Liu, Y.; Jiang, F. L. Au_xAg_{1-x} Nanocomposites with 40-Fold Emission Enhancement Formed by the Electrostatic Assembly of Gold Nanoclusters and Silver Nanoclusters for Bioimaging and Bioanalysis. *ACS Appl. Nano Mater.* **2019**, *2*, 408−417.
(32) Yin, M. M.; Chen, W. Q.; Lu, Y. Q.; Han, J. Y.; Liu, Y.; Jiang, F. L. A Model beyond Protein Corona: Thermodynamics and Stoichiometries of the Interactions between Ultrasmall Gold Nanoclusters and Proteins. *Nanoscale* **2020**, *12*, 4573−4585.
(33) Hedstrom, L. Serine Protease Mechanism and Specificity. *Chem. Rev.* **2002**, *102*, 4501−4524.
(34) David, M. B. Structure and Mechanism of Chymotrypsin. *Acc. Chem. Res.* **1976**, *9*, 145−152.
(35) Luo, Z.; Yuan, X.; Yu, Y.; Zhang, Q.; Leong, D. T.; Lee, J. Y.; Xie, J. From Aggregation-induced Emission of Au(I)-thiolate Complexes to Ultrabright Au (0) @ Au (I)-thiolate Core-shell Nanoclusters. *J. Am. Chem. Soc.* **2012**, *134*, 16662−16670.
(36) Shang, L.; Azadfar, N.; Stockmar, F.; Send, W.; Trouillet, V.; Bruns, M.; Gerthsen, D.; Nienhaus, G. U. One-pot Synthesis of Near-infrared Fluorescent Gold Clusters for Cellular Fluorescence Lifetime Imaging. *Small* **2011**, *7*, 2614−2620.
(37) Ghisaidoobe, A.; Chung, S. J. Intrinsic Tryptophan Fluorescence in the Detection and Analysis of Proteins: A Focus on Förster Resonance Energy Transfer Techniques. *Int. J. Mol. Sci.* **2014**, *15*, 22518−22538.
(38) Pandit, M. W.; Rao, M. S. N. Self-association of Proteins. Self-association of α-Chymotrypsin at Its Isoelectric Point in Buffer Solutions of Ionic Strength 0.1. *Biochemistry* **1975**, *14*, 4106−4110.
(39) Roux, S.; Garcia, B.; Bridot, J. L.; Salome, M.; Marquette, C.; Lemelle, L.; Gillet, P.; Blum, L.; Perriat, P.; Tillement, O. Synthesis, Characterization of Dihydrolipoic Acid Capped Gold Nanoparticles, and Functionalization by the Electroluminescent Luminol. *Langmuir* **2005**, *21*, 2526−2536.
(40) Yan, L.; Zhang, K.; Xu, H.; Ji, J.; Wang, Y.; Liu, B.; Yang, P. Target Induced Interfacial Self-assembly of Nanoparticles: A New Platform for Reproducible Quantification of Copper Ions. *Talanta* **2016**, *158*, 254−261.
(41) Shang, L.; Brandholt, S.; Stockmar, F.; Trouillet, V.; Bruns, M.; Nienhaus, G. U. Effect of Protein Adsorption on the Fluorescence of Ultrasmall Gold Nanoclusters. *Small* **2012**, *8*, 661−665.
(42) Perng, W.; Palui, G.; Wang, W.; Mattoussi, H. Elucidating the Role of Surface Coating in the Promotion or Prevention of Protein Corona around Quantum Dots. *Bioconjugate Chem.* **2019**, *30*, 2469−2480.
(43) Ross, P. D.; Subramanian, S. Thermodynamics of Protein Association Reactions: Forces Contributing to Stability. *Biochemistry* **1981**, *20*, 3096−3102.
(44) Singh, N.; Jabeen, T.; Sharma, S.; Roy, I.; Gupta, M. N.; Bilgrami, S.; Somvanshi, R. K.; Dey, S.; Perbandt, M.; Betzel, C.; Srinivasan, A.; Singh, T. P. Detection of Native Peptides as Potent Inhibitors of Enzymes Crystal Structure of the Complex Formed between Treated Bovine α-Chymotrypsin and an Autocatalytically Produced Fragment, Ile-Val-Asn-Gly-Glu-Glu-Ala-Val-Pro-Gly-Ser-Trp-Pro-Trp, at 2.2 A Resolution. *FEBS J.* **2005**, *272*, 562−572.
(45) Jurrus, E.; Engel, D.; Star, K.; Monson, K.; Brandi, J.; Felberg, L. E.; Brookes, D. H.; Wilson, L.; Chen, J.; Liles, K.; Chun, M.; Li, P.; Gohara, D. W.; Dolinsky, T.; Konecny, R.; Koes, D. R.; Nielsen, J. E.; Head-Gordon, T.; Geng, W.; Krasny, R.; Wei, G. W.; Holst, M. J.; McCammon, J. A.; Baker, N. A. Improvements to the APBS Biomolecular Solvation Software Suite. *Protein Sci.* **2018**, *27*, 112−128.
(46) Dolinsky, T. J.; Nielsen, J. E.; McCammon, J. A.; Baker, N. A. PDB2PQR: An Automated Pipeline for the Setup, Execution, and Analysis of Poisson-Boltzmann Electrostatics Calculations. *Nucleic Acids Res.* **2004**, *32*, 665−667.
(47) Stein, R. L.; Strimpler, A. M. Slow-binding Inhibition of Chymotrypsin and Cathepsin G by the Peptide Aldehyde Chymostatin. *Biochemistry* **1987**, *26*, 2611−2615.
(48) Birk, Y. The Bowman-Birk Inhibitor. *Int. J. Pept. Protein Res.* **1985**, *25*, 113−131.
(49) Ruan, J.; Yan, J.; Chen, H.; Cheng, J.; Sun, W.; Zhao, G. Purification and Properties of the Chymotrypsin Inhibitor from Wild Emmer Wheat (Triticum Dicoccoides) of Israel and Its Toxic Effect on Beet Armyworm, Spodoptera Exigua. *Pestic. Biochem. Physiol.* **2017**, *142*, 141−147.
(50) Bender, M. P. L. Inhibition of Serine Proteases by Arylboronic Acids. *Proc. Natl. Acad. Sci. U. S. A.* **1971**, *68*, 478−480.
(51) Iijima, K.; Katada, J.; Yasuda, E.; Uno, I.; Hayashi, Y. N-[2,2-Dimethyl-3-(N-(4-Cyanobenzoyl) Amino) Nonanoyl]-L-Phenylalanine Ethyl Ester as a Stable Ester-type Inhibitor of Chymotrypsin-like Serine Proteases: Structural Requirements for Potent Inhibition of α-Chymotrypsin. *J. Med. Chem.* **1999**, *42*, 312−23.
(52) Glazer, A. N. Inhibition of "Serine" Esterases by Phenylarsonic Acids. α-Chymotrypsin and the Subtilisins. *J. Biol. Chem.* **1968**, *243*, 3693−3701.
(53) André, L. L.; Ferreira, R. S.; Torquato, R. J. S.; Oliva, M. L. V.; Schuck, P.; Sousa, A. A. Allosteric Inhibition of α-Thrombin Enzymatic Activity with Ultrasmall Gold Nanoparticles. *Nanoscale Adv.* **2019**, *1*, 378−388.
(54) Ghosh, C.; Mondal, T.; Bhattacharyya, K. Enzyme Activity of α-Chymotrypsin: Deactivation by Gold Nano-cluster and Reactivation by Glutathione. *J. Colloid Interface Sci.* **2017**, *494*, 74−81.
(55) Bagchi, D.; Ghosh, A.; Singh, P.; Dutta, S.; Polley, N.; Althagafi, I. I.; Jassas, R. S.; Ahmed, S. A.; Pal, S. K. Allosteric Inhibitory Molecular Recognition of a Photochromic Dye by a Digestive Enzyme: Dihydroindolizine Makes α-Chymotrypsin Photo-responsive. *Sci. Rep.* **2016**, *6*, 34399−34409.
(56) Afosah, D. K.; Al-Horani, R. A.; Sankaranarayanan, N. V.; Desai, U. R. Potent, Selective, Allosteric Inhibition of Human Plasmin by Sulfated Non-saccharide Glycosaminoglycan Mimetics. *J. Med. Chem.* **2017**, *60*, 641−657.
(57) Henry, B. L.; Monien, B. H.; Bock, P. E.; Desai, U. R. A Novel Allosteric Pathway of Thrombin Inhibition: Exosite II Mediated Potent Inhibition of Thrombin by Chemo-enzymatic, Sulfated Dehydropolymers of 4-Hydroxycinnamic Acids. *J. Biol. Chem.* **2007**, *282*, 31891−31899.
(58) Verespy, S.; Mehta, A. Y.; Afosah, D.; Al-Horani, R. A.; Desai, U. R. Allosteric Partial Inhibition of Monomeric Proteases. Sulfated Coumarins Induce Regulation, not just Inhibition, of Thrombin. *Sci. Rep.* **2016**, *6*, 24043−24055.
(59) Afosah, D. K.; Verespy, S.; Al-Horani, R. A.; Boothello, R. S.; Karuturi, R.; Desai, U. R. A Small Group of Sulfated Benzofurans Induces Steady-state Submaximal Inhibition of Thrombin. *Bioorg. Med. Chem. Lett.* **2018**, *28*, 1101−1105.
(60) Zhao, D.; Zhou, J. Electrostatics-mediated α-Chymotrypsin Inhibition by Functionalized Single-walled Carbon Nanotubes. *Phys. Chem. Chem. Phys.* **2017**, *19*, 986−995.
(61) Sun, X.; Feng, Z.; Hou, T.; Li, Y. Mechanism of Graphene Oxide as an Enzyme Inhibitor from Molecular Dynamics Simulations. *ACS Appl. Mater. Interfaces* **2014**, *6*, 7153−7163.
(62) Zhang, X.; Bruning, J. B.; George, J. H.; Abell, A. D. A Mechanistic Study on the Inhibition of α-Chymotrypsin by a Macrocyclic Peptidomimetic Aldehyde. *Org. Biomol. Chem.* **2016**, *14*, 6970−6978.
(63) Yang, L.; Li, H.; Wu, P.; Mahal, A.; Xue, J.; Xu, L.; Wei, X. Dinghupeptins A-D, Chymotrypsin Inhibitory Cyclodepsipeptides Produced by a Soil-derived Streptomyces. *J. Nat. Prod.* **2018**, *81*, 1928−1936.
(64) Xiong, H.; Li, R. R.; Liu, S. Y.; Wu, F. X.; Yang, W. C.; Yang, G. F. Discovery of Specific Nonpeptide Probe for Chymotrypsin via Molecular Docking-based Virtual Screening and the Application. *ACS Appl. Bio Mater.* **2018**, *2*, 310−317.

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